

Full DHCP Network Configuration Guide

<https://gemini.google.com/app/b9489505e44aa38c>

User prompt: I wanna make the entire connection fully DHCP, so tell config for node pcs router and server so that all pcs are able to communicate within each other

Response: First, complete the physical wiring by connecting **Switch0 (Gig0/0)** to **Router0** and **Switch1 (Gig0/1)** to **Router0** using straight-through cables.

1. Router Configuration (Router0 CLI)

Configure the two gateway interfaces and set up the DHCP service on the router for the left subnet (Subnet A). For the right subnet (Subnet B), set up ip helper-address so the router forwards DHCP requests from PC3–PC5 to Server0.

```
Router> enable
Router# configure terminal
```

```
! Configure Left Subnet Interface (GigabitEthernet0/0)
Router(config)# interface GigabitEthernet0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
```

```
! Configure Right Subnet Interface (GigabitEthernet0/1)
Router(config)# interface GigabitEthernet0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
Router(config-if)# ip helper-address 192.168.2.10
Router(config-if)# no shutdown
Router(config-if)# exit
```

```
! Create DHCP Pool on Router for Subnet A (PC0 - PC2)
Router(config)# ip dhcp excluded-address 192.168.1.1
Router(config)# ip dhcp pool SUBNET_A_POOL
Router(dhcp-config)# network 192.168.1.0 255.255.255.0
Router(dhcp-config)# default-router 192.168.1.1
Router(dhcp-config)# exit
```

2. Dedicated Server Configuration (Server0)

Server0 provides DHCP for the right side (Subnet B) and requires a static IP address to function correctly as a server.

- **Static IP Settings (Desktop > IP Configuration):**

- **IP Address:** 192.168.2.10
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.2.1

- **DHCP Service Settings (Services > DHCP):**

- **Service:** On
- **Pool Name:** serverPool
- **Default Gateway:** 192.168.2.1
- **Start IP Address:** 192.168.2.11
- **Subnet Mask:** 255.255.255.0
- **Maximum number of Users:** 50
- Click **Save**.

3. Client Node Configuration (PC0 through PC5)

For every PC (PC0, PC1, PC2, PC3, PC4, PC5):

1. Click on the PC, navigate to the **Desktop** tab, and select **IP Configuration**.
2. Select **DHCP** (instead of Static).
3. Wait a few seconds for *DHCP request successful* to appear:
 - **PC0–PC2** will obtain IPs in the 192.168.1.X range (Gateway: 192.168.1.1).
 - **PC3–PC5** will obtain IPs in the 192.168.2.X range (Gateway: 192.168.2.1).

Once configured, test connectivity by sending a ping from **PC0** to **PC5** in the command prompt (ping 192.168.2.X).

User prompt: no cli and cant I choose auto cable?

Response: Yes, you can absolutely use **Auto Cable** (the lightning bolt icon) in Packet Tracer, and you don't need to touch the CLI for the router—you can configure it directly through its **GUI/Config tab**.

Here is how to set up the entire connection visually without writing CLI commands:

1. Complete the Cable Connections

- Select the **Connections** tool (lightning bolt icon on the bottom toolbar).
- Click the **Automatically Choose Connection Type** tool (the orange lightning bolt).
- Click **Switch0** and then **Router0**.
- Click **Router0** and then **Switch1**.

2. Router GUI Configuration (Router0)

Click on **Router0**, go to the **Config** tab, and configure both interfaces:

- **GigabitEthernet0/0 (Left Side Subnet A):**
 1. Click **GigabitEthernet0/0** under *Interfaces*.
 2. Check the **On** box for *Port Status*.
 3. Set **IP Configuration** to *Static*.
 4. **IP Address:** 192.168.1.1
 5. **Subnet Mask:** 255.255.255.0
- **GigabitEthernet0/1 (Right Side Subnet B):**
 1. Click **GigabitEthernet0/1** under *Interfaces*.
 2. Check the **On** box for *Port Status*.
 3. Set **IP Configuration** to *Static*.
 4. **IP Address:** 192.168.2.1
 5. **Subnet Mask:** 255.255.255.0

Note on Router DHCP without CLI: Packet Tracer's basic router GUI doesn't have visual fields to create full DHCP pools directly in the Config tab. To skip CLI entirely, use **Server0** as the single DHCP server for the whole network, or use a second server on the left side!

3. Set Up DHCP via Server0 (GUI Only)

Since Server0 is already on the right side, set it up to hand out IPs:

- **Set Server0 Static IP (Desktop > IP Configuration):**
 - **IP Address:** 192.168.2.10
 - **Subnet Mask:** 255.255.255.0
 - **Default Gateway:** 192.168.2.1
- **Enable DHCP Service (Services > DHCP):**
 1. Set **Service** to **On**.
 2. Set **Default Gateway:** 192.168.2.1
 3. Set **Start IP Address:** 192.168.2.11
 4. Set **Subnet Mask:** 255.255.255.0
 5. Click **Save**.

4. Client PC Configuration (PC0 to PC5)

For every PC (PC0 through PC5):

1. Click the PC, open the **Desktop** tab, and select **IP Configuration**.
2. Switch the radio button from **Static** to **DHCP**.

PC3, PC4, and PC5 will automatically grab IP addresses from Server0. For PC0–PC2 to grab IPs from Server0 across the router without CLI, you can simply add a second Server on Switch0 and turn on its DHCP service with 192.168.1.1 as the gateway!

User prompt: for pc0 DHCP request is failing and my faculty didnt ask to add extra server or cables

Response: The reason **PC0**'s DHCP request is failing is because PC0 sits on the left network, but your single DHCP server (**Server0**) sits on the right network across a router.

Routers naturally block broadcast traffic (including DHCP requests). To allow PC0–PC2 to get their IP addresses from **Server0** without adding extra hardware, you must tell **Router0** to relay those requests to Server0's IP address.

Step 1: Set Up Server0 (Must Have 2 Pools)

Open **Server0**, go to **Desktop > IP Configuration**, and assign it a static IP:

- **IP Address:** 192.168.2.10
- **Subnet Mask:** 255.255.255.0
- **Default Gateway:** 192.168.2.1

Next, go to **Services > DHCP** and create **two** pools so Server0 can issue IP addresses to *both* sides of the router:

- **Pool 1 (For Right Side - PC3 to PC5):**
 - **Pool Name:** serverPool (default)
 - **Default Gateway:** 192.168.2.1
 - **Start IP Address:** 192.168.2.11
 - **Subnet Mask:** 255.255.255.0
 - Click **Save**.
- **Pool 2 (For Left Side - PC0 to PC2):**
 - **Pool Name:** LeftNetwork
 - **Default Gateway:** 192.168.1.1

- **Start IP Address:** 192.168.1.10
- **Subnet Mask:** 255.255.255.0
- Click **Add**.

Step 2: Enable Router Interfaces & DHCP Relay

Open **Router0 CLI**. You only need to run these exact lines to turn on both interfaces and add the single required relay command (`ip helper-address`):

```
enable
configure terminal

! Left side configuration (Gig0/0)
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
 ip helper-address 192.168.2.10
 no shutdown
 exit

! Right side configuration (Gig0/1)
interface GigabitEthernet0/1
 ip address 192.168.2.1 255.255.255.0
 no shutdown
 exit
```

(Note: `ip helper-address 192.168.2.10` is what tells the router to take DHCP broadcast messages from the left side and forward them directly to Server0).

Step 3: Renew IP on PC0–PC5

1. Go to **PC0 > Desktop > IP Configuration**.
2. Toggle from **Static** to **DHCP** (or toggle back and forth if already selected).
3. PC0 will now successfully acquire an IP address in the 192.168.1.X range (such as 192.168.1.10), while PC3–PC5 will get IPs in the 192.168.2.X range.

User prompt: enable configure terminal ! Left side configuration (Gig0/0) interface GigabitEthernet0/0 ip address 192.168.1.1 255.255.255.0 ip helper-address 192.168.2.10 no shutdown exit ! Right side configuration (Gig0/1) interface GigabitEthernet0/1 ip address 192.168.2.1 255.255.255.0 no shutdown exit Cant this be done with gui

Response: No, `ip helper-address` cannot be set through the basic Packet Tracer GUI tab for Cisco 1941 routers.

Cisco Packet Tracer's GUI tab under **Config > Interfaces** only provides basic fields for:

- Port Status (On/Off)

- IP Address
- Subnet Mask

It lacks fields for advanced interface commands like DHCP relay services (`ip helper-address`).

The Alternative: Full GUI Setup (Using Router as DHCP Server)

If your faculty insists on **100% GUI without typing CLI commands**, you must configure **Router0 to handle DHCP directly for the left side** instead of using Server0 across subnets.

Here is how to do it entirely via GUI:

1. Configure Router Interfaces (Router0 > Config)

- **GigabitEthernet0/0:**
 - Port Status: **On**
 - IP Address: 192.168.1.1 | Subnet Mask: 255.255.255.0
- **GigabitEthernet0/1:**
 - Port Status: **On**
 - IP Address: 192.168.2.1 | Subnet Mask: 255.255.255.0

2. Configure Server0 (Right Subnet DHCP)

- Set Static IP: 192.168.2.10 / 255.255.255.0 (Gateway: 192.168.2.1)
- Go to **Services > DHCP:**
 - Service: **On**
 - Default Gateway: 192.168.2.1
 - Start IP: 192.168.2.11
 - Subnet Mask: 255.255.255.0
 - Click **Save**.

3. Configure Router's Built-in DHCP (Left Subnet DHCP) Instead of relaying to Server0, let Router0 act as the local DHCP server for the left side:

- Open **Router0 > Config** tab.
- Look at the left sidebar under **SERVICES** and click **DHCP**.
- Set **Service: On**.
- **Pool Name:** DEFAULT-POOL (or create a new pool)

- **Default Gateway:** 192.168.1.1
- **Start IP Address:** 192.168.1.10
- **Subnet Mask:** 255.255.255.0
- Click **Save**.

Summary

- **If Server0 must give IPs to both sides:** You **must** paste those 3 CLI lines (`ip helper-address 192.168.2.10`) into the router's CLI tab.
 - **If you want zero CLI:** Turn on the **DHCP Service directly inside Router0's GUI** for the left side, and use **Server0's GUI** for the right side.
-

Response: You are getting that error because you were already inside interface configuration mode (`Router(config-if)#`) when typing `enable` and `configure terminal`. Those commands only work from user/privilege mode.

Since your CLI prompt currently shows `Router(config-if)#`, you are already in the right mode to add the `ip helper-address`!

Just copy and paste these exact lines directly into the CLI window:

```
exit
interface GigabitEthernet0/0
ip helper-address 192.168.2.10
exit
```

Why this works:

1. Typing `exit` takes you out of `GigabitEthernet0/1` and back to global config (`Router(config)#`).
2. Entering `interface GigabitEthernet0/0` lets you target the left network interface.
3. The `ip helper-address 192.168.2.10` command tells the router to pass DHCP requests from PC0–PC2 over to Server0.

After running this, go to **PC0 > Desktop > IP Configuration**, switch it to **Static**, and then back to **DHCP**. It will fetch an IP successfully.

User prompt: No I meant there's nothing for DHCP in Router0

Response: Ah, I see what you're looking at! Yes, on actual Cisco routers (like the 1941 shown in your topology), Packet Tracer's GUI **Config** tab does not have a "DHCP" option in the left sidebar under SERVICES. Only generic server devices display that GUI option.

Because the router GUI lacks a visual DHCP button, you have **two ways** to solve this:

Option 1: Use Server0 for Everything (Recommended by Faculty)

Keep the single Server0 handling DHCP for the whole network. You just need to apply the relay command on the router via the **CLI tab** (not the bottom "Equivalent IOS Commands" box, but the actual **CLI** tab at the top).

1. Click the **CLI** tab at the top of the Router0 window.
2. Press **Enter** a couple of times until you see a prompt like Router# or Router(config-if)#.
3. Type exit or end until you reach Router(config)#.
4. Run these lines:

```
interface GigabitEthernet0/0
ip helper-address 192.168.2.10
exit
```

(This tells the router to pass PC0–PC2's DHCP requests across to Server0).

Option 2: Turn Router0 into the DHCP Server via CLI

If your faculty wants Router0 to directly issue IP addresses to PC0–PC2 without sending requests to Server0, configure the router's internal DHCP pool in the **CLI** tab:

```
enable
configure terminal
ip dhcp excluded-address 192.168.1.1
ip dhcp pool LEFT_LEGS
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
exit
```

Once you enter either set of commands in the CLI tab, toggle **PC0** from **Static** -> **DHCP**, and it will grab its IP address immediately!
